

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Naveen

May 2024

Course Code: CIV102

Course: ENGINEERING MECHANICS

QP No. :U177-2

Duration: 3 hours

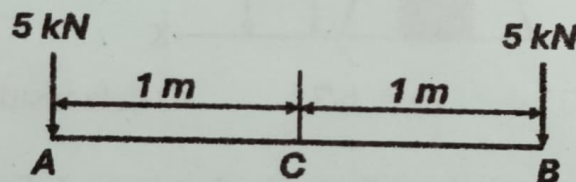
Max. Marks:100

PART – A

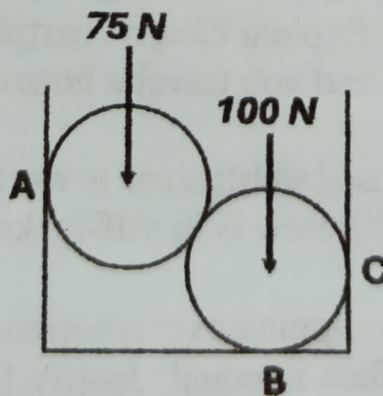
Answer all the questions

10 x 2 = 20 Marks

1. Determine the equilibrant and its location for the force system shown in below Figure.

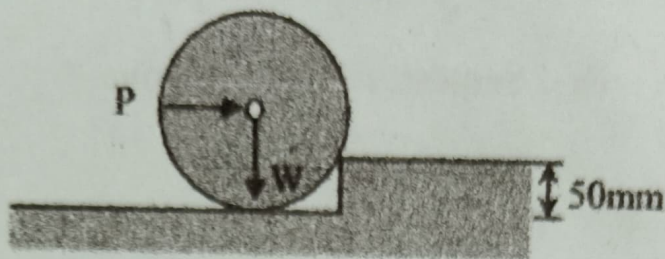


2. Two cylinders of radius 100 mm and 120 mm weighing 75 N and 100 N respectively are placed in an open rectangular channel of width 350 mm as shown in Fig. Determine the reaction force at point B.

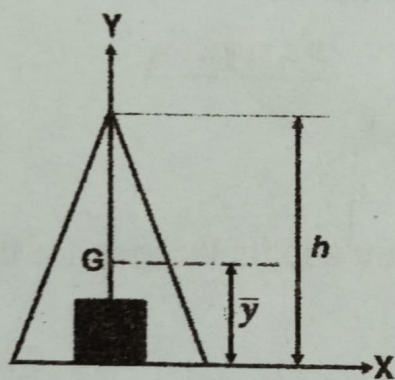


3. A cylinder of radius 250 mm and weight, $W=10$ kN is rolled up an obstacle of height 50 mm by applying a horizontal force P at its centre

as shown in the figure. All interfaces are assumed frictionless. The minimum value of P is _____.



4. The shaded portion has been carved out of the isosceles triangle of height ' h '. The centroid of the remaining area will be at a distance from the base of the triangle that is _____.



- a) $> 'h/3'$ b) $< 'h/3'$ c) $= 'h/3'$ d) data insufficient.

5. What are the characteristics of a body that help define its Moment of Inertia?
6. You would do more work trying to carry a 5 kg mass over a horizontal distance of 10 m than a mass of 1 kg over the same distance by walking. True or false? Explain clearly, particularly as the weight of the mass acts vertically and you travel a horizontal distance.
7. A screw jack which is used to lift a car is working at an efficiency of 60%. Check whether the screw is in self-locking condition or not?
8. A cylinder rolls without slipping over a horizontal plane. The velocity at all points on its surface is equal. Justify the sentence is True or False.

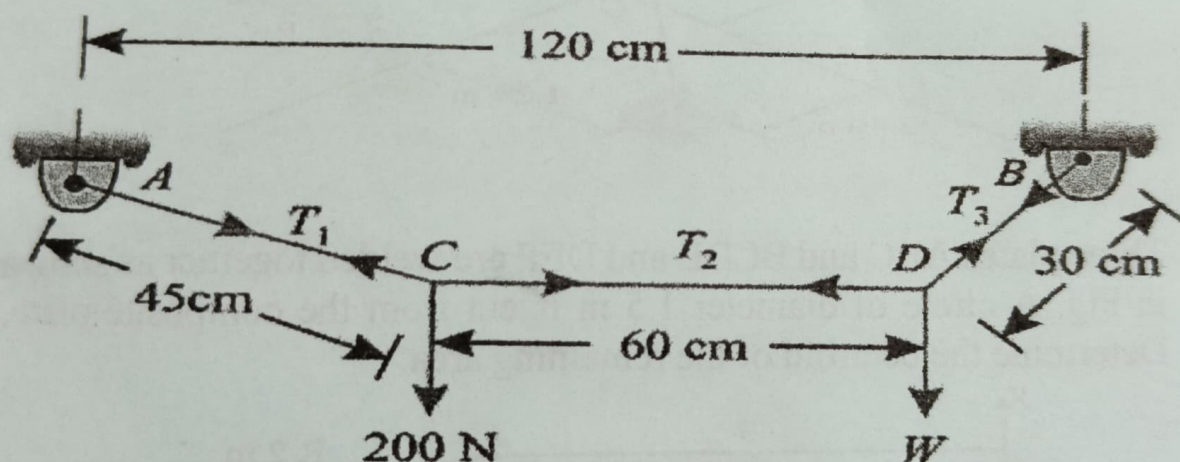
9. The motion of a particle is described by the relation $s = 9t^2 + 10t + 5$, where 't' is time in seconds. Determine the initial velocity and acceleration.
10. How is "impulse" related to "momentum" in dynamics?

PART - B

Answer all the questions

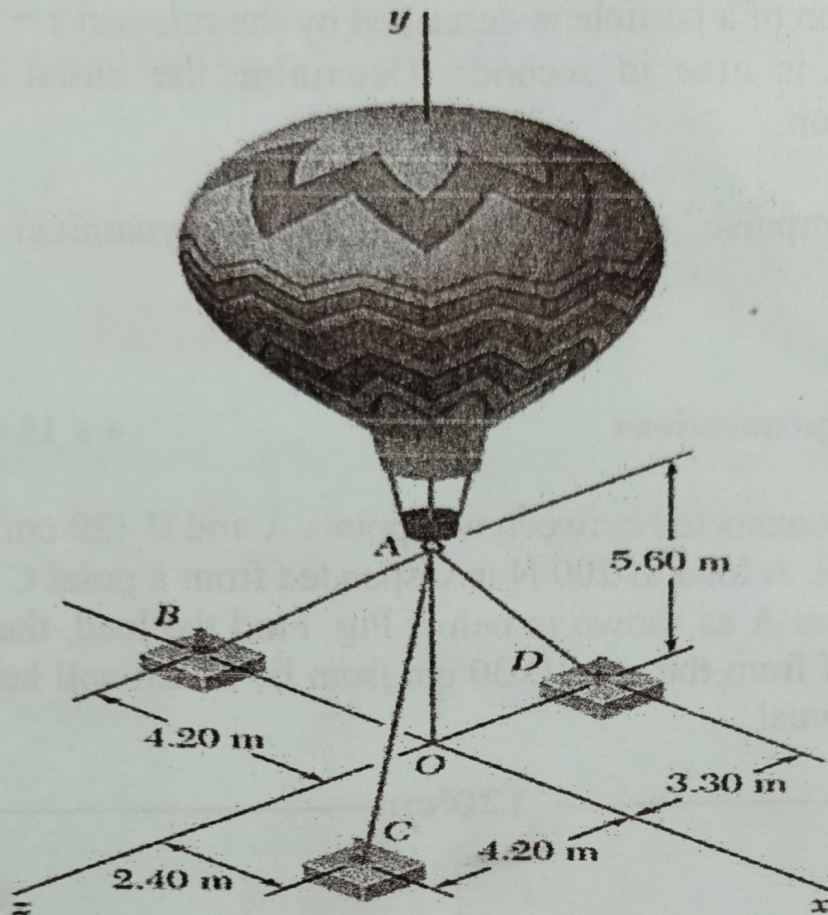
4 x 15 = 60 Marks

11. A rope is connected between two points A and B 120 cm apart at the same level. A load of 200 N is suspended from a point C on the rope 45 cm from A as shown in below Fig. Find the load, that should be suspended from the rope D 30 cm from B, which will keep the rope CD horizontal.

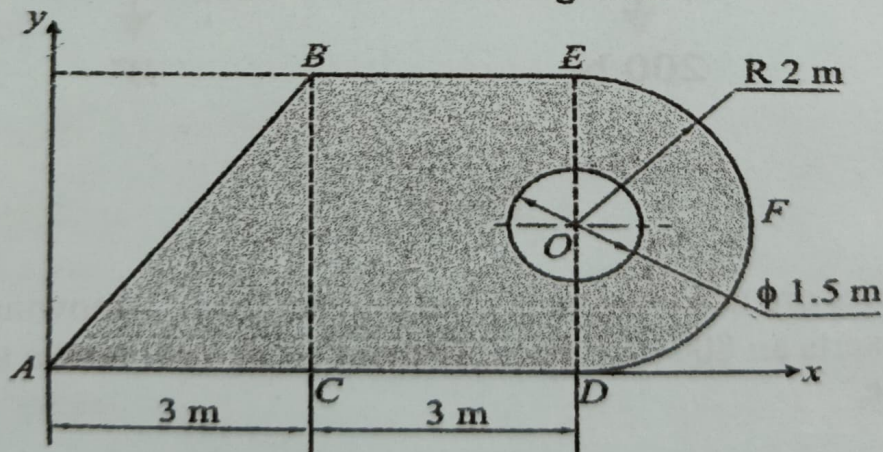


(OR)

12. Three cables are used to tether a balloon as shown. Knowing that the balloon exerts an 800-N vertical force at A, determine the tension in each cable.

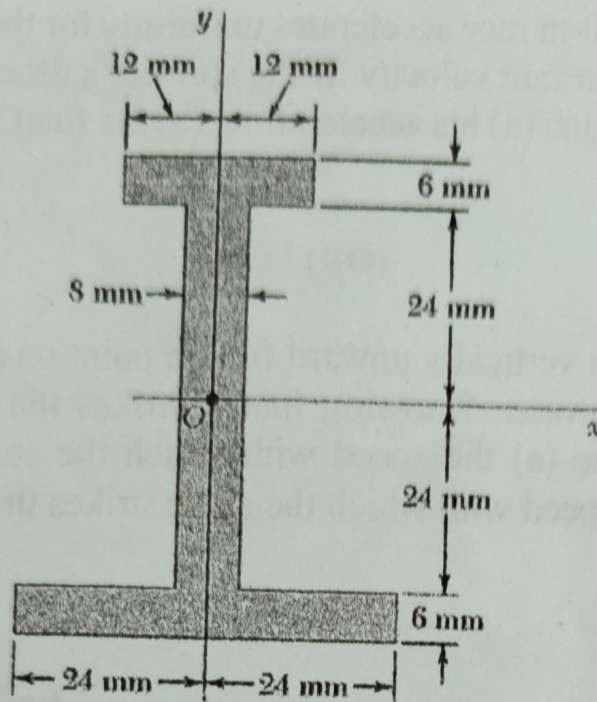


13. Three plates ABC and BCDE and DEF are welded together as shown in Fig. A circle of diameter 1.5 m is cut from the composite plate. Determine the centroid of the remaining area.

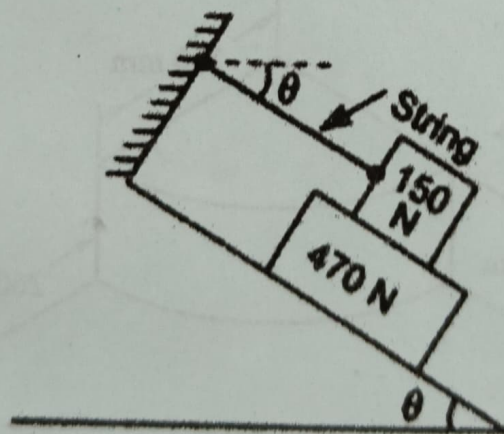


(OR)

14. Determine the moment of inertia and the radius of gyration of the shaded area with respect to the x axis.



15. The inclined plane shown in Fig. supports two blocks of weights 150 N and 470 N. Determine the angle of inclination θ of the plane so that the motion of the heavier block is impending down the plane. What is the string tension for this condition? The co-efficient of friction for all surfaces is $1/3$.



(OR)

16. A uniform ladder of 4 m length rests against a vertical wall with which it makes an angle of 45° . The coefficient of friction between the ladder and the wall is 0.4 and that between ladder and the floor is 0.5. If a man, whose weight is one-half of that of the ladder ascends it, how high will it be when the ladder slips?

17. A sprinter in a 100-m race accelerates uniformly for the first 35 m and then runs with constant velocity. If the sprinter's time for the first 35 m is 5.4 s, determine (a) his acceleration, (b) his final velocity, (c) his time for the race.

(OR)

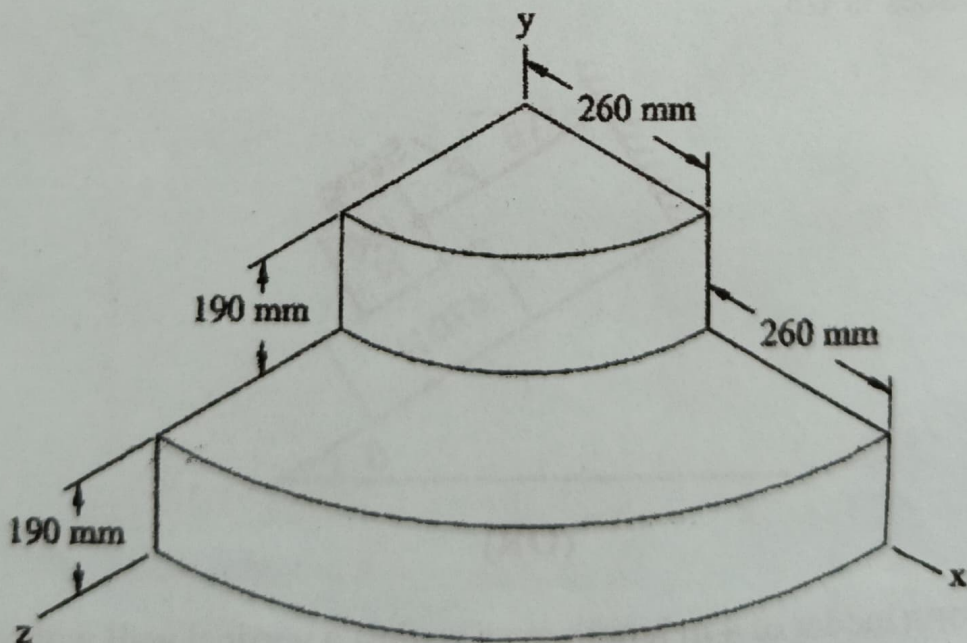
18. A stone is thrown vertically upward from a point on a bridge located 40 m above the water. Knowing that it strikes the water 4 s after release, determine (a) the speed with which the stone was thrown upward, (b) the speed with which the stone strikes the water.

PART - C

Answer the following

1 x 20 = 20 Marks

19. a) The concrete steps shown are in the shape of a quarter circle. If the centroidal distance from the axis of revolution is 216.67 mm, Determine the cubic meters of concrete required to construct the steps. (6)



- b) Find the acceleration and tension in the string of the system shown in Fig. Co-efficient of friction can be taken as 0.3 for all planes of contact. Pulley is smooth. Also determine the velocity of the system in 4 seconds after starting from rest. (14)

